

JACOB COOK

Security Engineer (SEAT / SIRT) | Detection Engineering & IR | Sigma · MITRE ATT&CK · Python · Cloud Forensics · NIST/SANS Lifecycle

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Detection-focused security engineer with hands-on experience triaging and responding to security events, authoring Sigma rules mapped to MITRE ATT&CK, and partnering on detection tuning to reduce noise. Strong in Python and Bash automation, on-call discipline, and structured incident write-ups for technical and non-technical audiences. Comfortable with the SANS/NIST IR lifecycle, runbook authoring, and post-incident metrics. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress.

TECHNICAL SKILLS

Detection & IR: Escalated Event Response, Cyber Kill Chain, MITRE ATT&CK / D3FEND, NIST & SANS IR Lifecycle, Host/Network/Cloud Investigation Concepts, Root Cause & Timeline Reporting, Runbook & Playbook Authoring, On-Call Coverage

SIEM, EDR & Tooling: Wazuh SIEM (build & admin), Microsoft Sentinel (familiar), Defender (familiar), KQL/SPL (familiar), EDR Concepts, OWASP, CIS, OSINT, AI-Augmented Investigation

Cloud & Identity: Azure AD / Entra ID, Okta, AWS labs (IAM, Network Security), SSO / RBAC, Terraform

Scripting: Python, Bash, PowerShell (familiar), SQL, FastAPI, Git

TECHNICAL PROJECTS

Stryker / Intune Detection Pack: Detection-as-Code

2026

- Authored 6 Sigma-format detection rules mapped to MITRE ATT&CK / D3FEND that eliminated 100% of false positives in a 100-event simulation — direct partnership-with-detection-engineering pattern from incident lessons learned to shipped detection.
- Documented 18 benign false-positive scenarios as repeatable evidence — the same documentation discipline post-incident reviews drive.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py).

Blue Team SOC Monitoring Lab: Wazuh SIEM

2026

- Built a full Wazuh SIEM stack from scratch and validated 12 detection patterns against 242 simulated events with 100% precision after tuning — direct SIEM/EDR-style log analysis and correlation work.
- Reduced false-positive volume by 100% by tuning out 79 benign-only matches without dropping a single true positive.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).

SOAR-lite: Incident Response Orchestrator

2026

- Built end-to-end IR automation in Python/FastAPI: webhook ingestion, playbook routing, IP blocking, Slack notification, and incident write-up — automating response actions from incident lessons learned.
- Reduced triage time by over 60% in lab testing while preserving full evidence trail per incident.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py).

Phishing Analysis Lab

2026

- Triageed 5 real-world phishing samples end-to-end with header inspection, SPF/DKIM/DMARC validation, IOC extraction, and attacker infrastructure mapping — 100% accurate verdicts against ground truth.
- Delivered timeline reconstruction and IOC trail in NIST-style incident format.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6)

2024

- Engineered RBAC and Azure AD SSO controls with documented data-privacy boundaries — direct cloud IAM and security-control implementation work.
- Shipped enterprise-grade platform on deadline under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response

2019 - 2025

- Maintained 99.5% system uptime by triaging telemetry alerts on rotating shifts — six years operating in an on-call coverage model with high-pressure decision-making.
- Resolved 200+ anomalies through structured root cause analysis with tracked documentation that reduced repeat incidents by an estimated 30% — direct post-incident metrics discipline.
- Reduced mean time to recovery by an estimated 20% via cross-team coordination, escalation, and post-incident reporting.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress)

Expected Oct 2026 · GPA 3.50

CSUS | BS, Computer Science

Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)